

**VIT**Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

REG.NO.:

**SCHOOL OF ELECTRONICS ENGINEERING
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025**

SLOT: C1 +TC1+TCC1

Programme Name & Branch : B. Tech BVD
Course Code and Course Name : BAECE103 – Network Theory
Faculty Name(s) : Dr.Elizabeth Rufus, Dr. Sathya P
Class Number(s) : VL2025260105990, VL2025260105992
Date of Examination : 07.10.2025
Exam Duration : 90 minutes **Maximum Marks: 50**

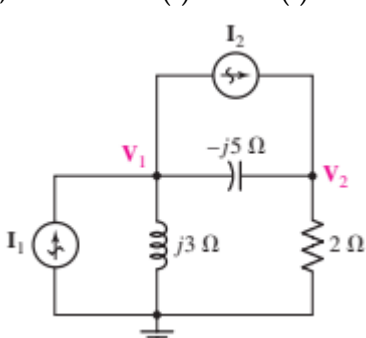
General instruction(s):

- **Open Book Test - Answer All Questions.**
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes:

CO2: Determine the natural and step responses of RC, RL, and RLC circuits.

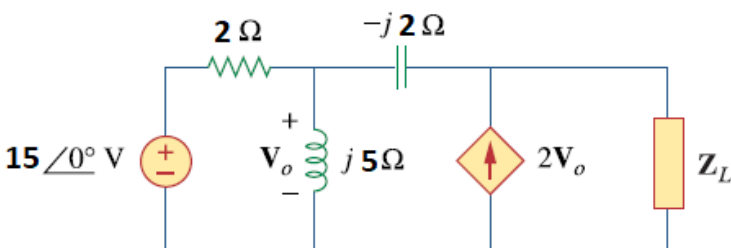
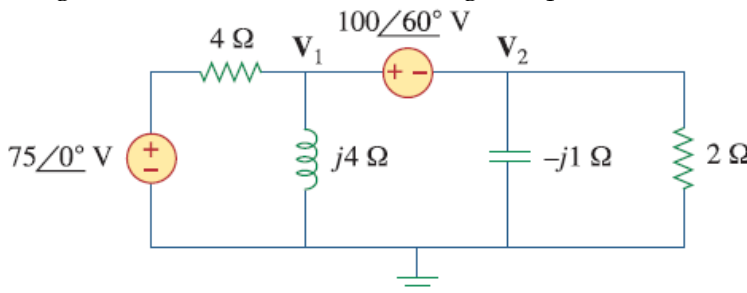
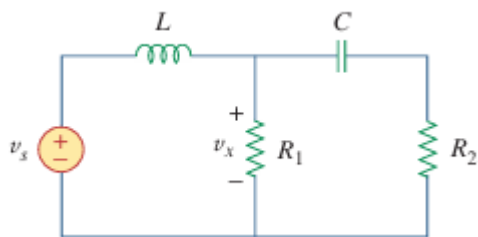
CO3: Apply appropriate techniques and theorems for sinusoidal steady-state analysis.

CO4: Apply the Laplace transform to circuits for steady-state and transient analysis.

Q. No	Question	M	CO	BL
1.	If a parallel RLC circuit is constructed from component values $C = 16 \text{ mF}$ and $L = 1 \text{ mH}$, choose R such that the circuit is (a) just barely overdamped; (b) just barely underdamped; (c) critically damped. (d) Does your answer for part (a) change if the resistor tolerance is 1%? 10%? Justify (e) Increase the exponential damping coefficient (ratio α/ω_0) for part (c) by 20%. Is the circuit now underdamped, overdamped, or still critically damped? Explain.	10	2	3
2.	The phasor domain circuit of Fig.1 was drawn assuming an operating frequency of 2.5 rad/s. Unfortunately, the manufacturing unit installed the wrong sources, each operating at a different frequency. If $i_1(t) = 4 \cos 40t \text{ mA}$ and $i_2(t) = 4 \sin 30t \text{ mA}$, calculate $v_1(t)$ and $v_2(t)$.  Fig.1	10	3	3

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3.	In the circuit shown in Fig.2, find the value of Z_L that will absorb the maximum power and the value of the maximum power delivered to the load.  Fig.2	10	3	3
4.	Find the Norton's equivalent circuit and calculate the current passing through the 2Ω resistor and the voltage drop across the same resistor.  Fig.3	10	3	3
5.	a) Find $F(s)$ if: $f(t) = 8e^{-3t} \cosh tu(t - 2)$ (5 marks) b) Find v_x in the circuit shown in Fig.4 using Laplace Transform. $R_1 = 2\Omega$, $R_2 = 4\Omega$, $L = 1\text{ H}$, $C = 1/8\text{ F}$ and $v_s(t) = 4u(t)\text{ V}$. (5 marks)  Fig.4	10	4	3

(1)

CAT-2

Slot C1

Key

1. Given $C = 16 \mu\text{F}$ $L = 1 \text{ mH}$.

a) overdamped. $\alpha = \frac{1}{2RC}$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-3} \times 16 \times 10^{-6}}} = 250 \text{ rad/s}$$

$$R < \frac{1}{2} \sqrt{\frac{L}{C}} \Rightarrow R < 0.125$$

$$\therefore \text{Let } R = 0.124 \Omega$$

$$\text{now } \alpha = 252 \text{ s}^{-1} \text{ and } \alpha > \omega_0$$

b) $\alpha < \omega_0$. $R > 0.125$ underdamped.

$$\therefore \text{Let } R = 0.126 \Omega$$

$$\text{now } \alpha = 248 \text{ s}^{-1} \text{ and } \alpha < \omega_0$$

c) critically damped

$$\alpha = \omega_0$$

$$\text{then } R = 0.125 \Omega$$

d) If the resistor tolerance is $\pm 1\%$ in your part.

$$1\% \Rightarrow \frac{1}{100} \times 0.124 = 1.24 \times 10^{-3}$$

$$R_{\text{new}} = 0.124 + 1.24 \times 10^{-3}$$

$$R_{\text{new}} = 0.12524$$

$$\text{then } \alpha = 249.52 \text{ causing underdamping. } (< \omega_0)$$

$$\text{Now } R_{\text{new}} = 0.124 - 1.24 \times 10^{-3}$$

$$R_{\text{new}} = 0.12276$$

$$\text{causing } \alpha = 254.56 \text{ this is } > \omega_0 - \text{overdamping}$$

11% ω_0 . $\alpha = 229.1 < \omega_0$ - underdamped

$$\alpha = 280 > \omega_0 - \text{overdamped.}$$

e. If $\frac{\alpha}{\omega_0} = \text{d\%}$. $\uparrow$ that is earlier $\frac{\alpha}{\omega_0} = 1$

$$\text{now } \frac{\alpha}{\omega_0} = 1.2 = \frac{\alpha}{250}$$

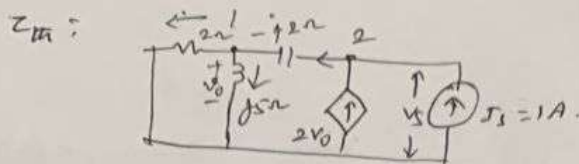
$$\Rightarrow \alpha = 300 \text{ s}^{-1}$$

$\therefore \alpha > \omega_0$
overdamped
when damping
coeff $\uparrow$ this
happens.

C1 - (2)

2. $v_1(t) = 7.995 \cos(40t + 2.7^\circ) + 0.343 \cos(30t + 90.1^\circ) \text{ mV}$
 $v_2(t) = 7.995 \cos(40t + 2.4^\circ) + 1.67 \cos(30t - 180^\circ) \text{ mV}$

3.



node 2:

$$2V_0 + 1 = \frac{V_3 - V_0}{-j2} = 0.5j(V_3 - V_0)$$

$$2V_0 + j0.5V_0 = j0.5V_3 - 1$$

$$V_0(2 + j0.5) = j0.5V_3 - 1 \quad \text{--- (2)}$$

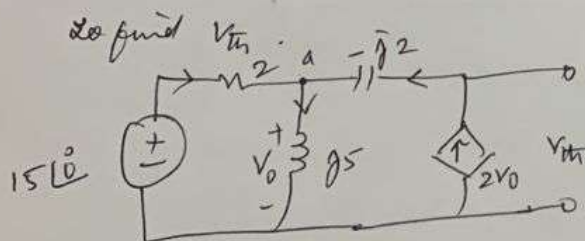
solving $V_0 = 0.6608 \angle -7.595^\circ \text{ V} = 0.655 - j0.087 \text{ V}$

$$\textcircled{1} \Rightarrow V_0(0.5 + j0.3) = j0.5V_3$$

$$V_3 = \frac{V_0(0.5 + j0.3)}{j0.5} = 0.7706 \angle -66.63^\circ$$

$$Z_{th} = \frac{V_0}{I_s} = 0.7706 \angle -66.63^\circ = 0.306 - j0.707 \Omega$$

$$Z_L = Z_{th}^*$$



kcl at node a

$$\frac{15 - V_0}{2} + 2V_0 = \frac{V_0}{j5}$$

$$V_0 = \frac{15 \angle 180^\circ}{3.0265 \angle 7.594^\circ} = 4.956 \angle 172.4^\circ$$

kvl to outer loop:

$$-V_0 - j2I + V_{th} = 0$$

$$V_{th} = V_0 + j2(-2V_0)$$

$$V_{th} = V_0(1 - j4) = (5 \angle 172.4^\circ)(4.123 \angle -75.963^\circ)$$

$$V_{th} = 20.615 \angle 96.04^\circ \text{ V}$$

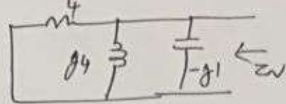
$$P_{max} = \frac{V_{th}^2}{8R_{th}} = \frac{(20.615)^2}{8 \times 0.306} = 173.6 \text{ W}$$

$$P_{max} = 173.6 \text{ W}$$

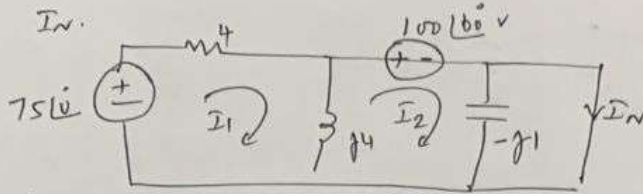
4) Norton's equivalent.

(3)

Z_n



$$\begin{aligned} Z_n &= (4 \parallel j4) \parallel (-j1) \\ &= (2 + j2) \parallel (-j1) \\ &= 1.2647 \angle -71.56^\circ = 0.4 - j1.2 \Omega \end{aligned}$$



Loop 2

$$\begin{aligned} -75 + 4I_1 + j4(I_1 - I_2) &= 0 \\ 4I_1 + j4(I_1 - I_2) &= 75 \\ 4(I_1 + j4I_1 - j4I_2) &= 75 \\ I_1(4 + j4) - j4I_2 &= 75 \quad \text{--- (1)} \end{aligned}$$

Loop 2

$$\begin{aligned} j4(I_2 - I_1) + 100\angle 0^\circ &= 0 \\ -j4I_1 + j4I_2 &= -100\angle 0^\circ \quad \text{--- (2)} \end{aligned}$$

$$I_2 = I_n. \quad I_2 = \frac{286.62 \angle -59.28^\circ}{16 \angle 90^\circ} = 17.91 \angle 210.72^\circ \text{ A}$$



$$\begin{aligned} I_0 &= 17.91 \angle 210.72^\circ \times \frac{0.4 - j1.2}{0.4 - j1.2 + 2} \\ &= (0.47 \angle -45.01^\circ) (17.91 \angle 210.72^\circ) \end{aligned}$$

$$I_0 = 8.42 \angle 165.71^\circ \text{ A}$$

$$V_0 = I_0 \times 2 \Omega = 16.84 \angle 165.71^\circ \text{ V rms.}$$

C1 - (4)

5. a) $F(s) = ?$ $f(t) = 8e^{-3t} \cosh t u(t-2)$.

$$\cosh t = \frac{e^t + e^{-t}}{2}$$

$$L[f(t-a)u(t-a)] = e^{-as} F(s)$$

$$f(t) = 8e^{-3t} \cdot \frac{1}{2} (e^t + e^{-t}) \cdot u(t-2)$$

$$f(t) = 4e^{-3t} (e^t + e^{-t}) u(t-2)$$

$$= 4e^{-2t} u(t-2) + 4e^{-4t} u(t-2)$$

$$L[f(t)] = L\left[4e^{-4} e^{-2(t-2)} u(t-2) + 4e^{-8} e^{-4(t-2)} u(t-2)\right]$$

first term $= L\left[4e^{-4} e^{-2(t-2)} u(t-2)\right] = 4e^{-4} e^{-2s} \cdot L[e^{-2t} u(t)]$

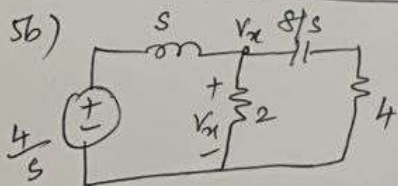
$$= 4e^{-4} e^{-(2s+4)} \cdot \frac{1}{s+2}$$

$$= \frac{4e^{-(2s+4)}}{s+2}$$

second term $= L\left[4e^{-8} e^{-4(t-2)} u(t-2)\right] = \frac{4e^{-(2s+8)}}{s+4}$

$$\therefore f(s) = \frac{4e^{-(2s+4)}}{s+2} + \frac{4e^{-(2s+8)}}{s+4}$$

$$F(s) = \frac{e^{-(2s+4)} [(4e^2 + 4e^{-2})s + (16e^2 + 8e^{-2})]}{s^2 + 6s + 8}$$



$$\frac{V_x - \frac{4}{s}}{s} + \frac{V_x}{2} + \frac{V_x}{4 + \frac{8}{s}} = 0$$

$$\frac{V_x - \frac{4}{s}}{s} + \frac{V_x}{2} + \frac{V_x s}{4s + 8} = 0$$

$$V_x \left(\frac{1}{s} + \frac{1}{2} + \frac{s}{4s+8} \right) = \frac{4}{s^2}$$

$$V_x \Rightarrow \frac{4}{s^2} = \frac{2(4s+8) + s(4s+8) + 2s^2}{2s(4s+8)} \cdot V_x$$

$$V_n = \frac{4}{s^2} \frac{(2s)(4s+8)}{8s+16+4s^2+8s+2s^2}$$

(5)

$$= \frac{64s^2 + 64s}{8s^3 + 16s^2 + 4s^4 + 8s^3 + 2s^4} = \frac{64s(1+s)}{6s^4 + 16s^3 + 16s^2}$$

$$= \frac{64(1+s)}{6s^3 + 16s^2 + 16s} = \frac{64(1+s)}{8 \cdot (3s^3 + 8s^2 + 8s)}$$

$$= \frac{32(1+s)}{s(3s^2 + 8s + 8)}$$

$$= \frac{A}{s} + \frac{Bs+C}{3s^2+8s+8}$$

$$A(3s^2+8s+8) + Bs^2 + Cs = 32 + 32s$$

$$3A + B = 0 \Rightarrow B = -3A$$

$$8A + C = 32 \Rightarrow 32 + C = 32$$

$$8A = 32$$

$$C = 0$$

$$A = 4$$

$$B = -3A$$

$$B = -12$$

$$\therefore V_n = \frac{4}{s} - \frac{12s}{3s^2+8s+8}$$

$$= \frac{4}{s} - \frac{12s}{(s+s_1)(s+s_2)}$$

$$= \frac{4}{s} - \frac{12s}{s+s_1} - \frac{12s}{s+s_2}$$

$$= \frac{4}{s} - \frac{12s}{s - (1.33 - j0.944)} - \frac{12s}{s - (1.33 + j0.944)}$$

$$= 4u(t) - 12 \cos(\omega t)$$

$$s_{1,2} = \frac{-8 \pm \sqrt{64 - 96}}{6}$$

$$= \frac{-8 \pm \sqrt{-32}}{6}$$

$$= \frac{-8 \pm \sqrt{(-1)32}}{6}$$

$$s_1 = \frac{-8 + j5.656}{6}$$

$$s_2 = \frac{-8 - j5.656}{6}$$

$$s_{1,2} = -1.33 \pm j0.944$$